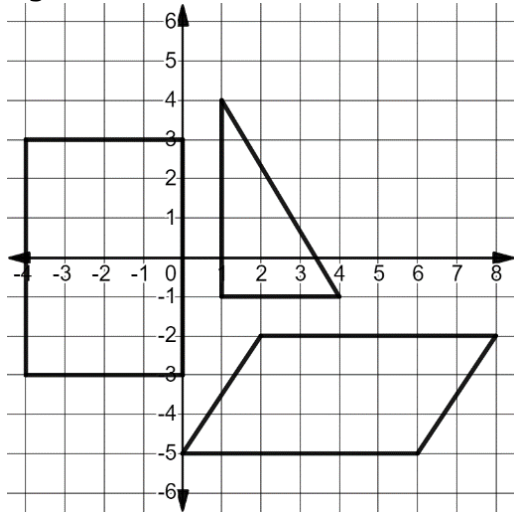


Geometry
Unit 9
Lesson 1 Homework

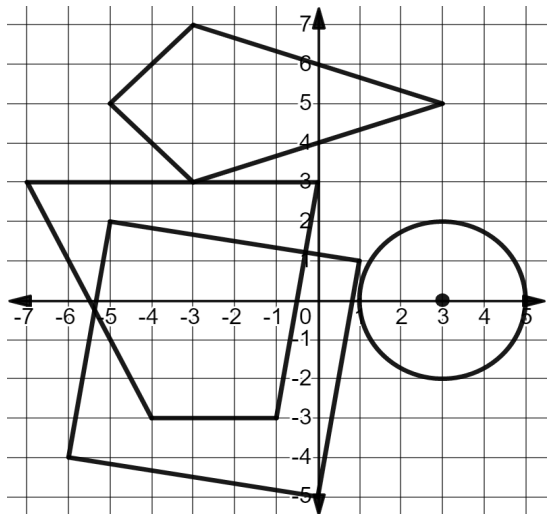
Name _____ **Answer Key** _____
Date _____
Period _____

A graph is shown with a rectangle, triangle, and parallelogram. Find the area of each figure.



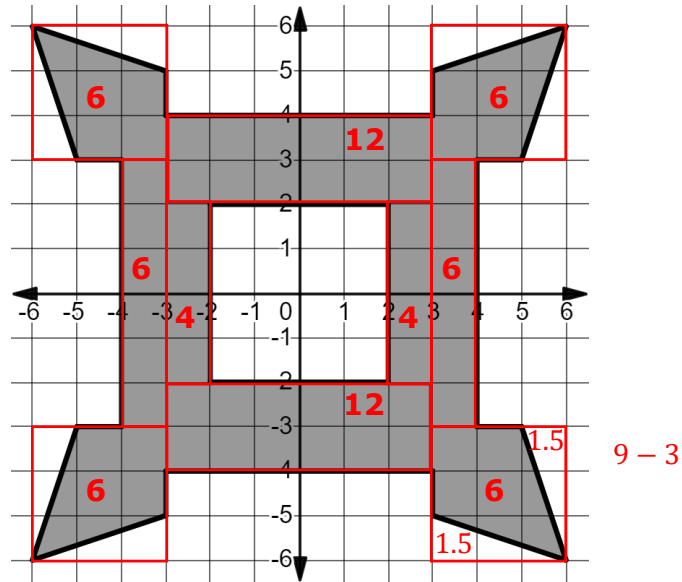
Shape	Area
Rectangle	1. $6 \times 4 = 24$ square units
Triangle	2. $(3 \times 5) \frac{1}{2} = \frac{15}{2} = 7.5$ units ²
Parallelogram	3. $3 \times 6 = 18$ units ²

A graph with a trapezoid, kite, circle, and rhombus is shown. Find the area of each figure.



Shape	Area
Kite	4. $(4 \times 8) \frac{1}{2} = 16$ units ²
Trapezoid	5. $\frac{1}{2} \cdot (6)(3 + 7) = 30$ units ²
Rhombus	6. $\frac{\sqrt{74} \cdot \sqrt{74}}{2} = 37$ units ²
Circle	7. $4\pi \approx 12.56$ units ²

A graph similar to the outline of the Castillo de San Marcos in St. Augustine, Florida is shown.



8. What shapes can be used to find the area of the shaded region?

Answers will vary.

Possible Answers: triangles, squares, rectangles

9. Divide the shaded area into smaller shapes on the graph.

10. What is the area of the shaded region?

68 square units

The table includes the populations of a few cities in Florida.

City	Population in 2014	Population in 2019
Destin	12,840	13,702
Panama City	36,405	36,640
Pensacola	52,505	52,642

1. Pensacola is spread over an area of 41.12 square miles. What was the population density in 2014?

$$\frac{52505}{41.12} \approx 1277 \frac{\text{people}}{\text{sq. mile}}$$

2. What was the population density in 2019?

$$\frac{52642}{41.12} \approx 1280 \frac{\text{people}}{\text{sq. mile}}$$



3. How did the population density change over the course of 5 years?

It increased by about 3 people per square mile.

4. Destin is spread over an area of 7.66 square miles and Panama City covers 41.27 square miles. What was the population density of Destin in 2019?

$$\frac{13702}{7.66} \approx 1789 \frac{\text{people}}{\text{sq. mile}}$$

5. What was the population density of Panama City in 2019?

$$\frac{36640}{41.27} \approx 888 \frac{\text{people}}{\text{sq. mile}}$$

6. How does the population density of Destin compare with the population density of Panama City?

Destin has about 901 more people per square mile.

7. Manatee Springs is one natural spring that provides warm water for manatees to survive. The spring run that allows water to flow from the spring into the Suwannee River is $\frac{1}{4}$ mile long. There are about 20 manatees that can be seen regularly here. If the spring run is $\frac{1}{4}$ mile wide, what is the population density of the manatees?

$$\frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$$

$$\frac{20}{\frac{1}{16}} = 320 \text{ manatees/sq. mile}$$

8. Miami is a popular tourist spot due to its beaches and warm weather. It spreads across an area of 55.25 square miles. If 24.2 million people visit the area, what is the density of visitors?

$$\frac{24.2 \text{ million}}{55.25 \text{ sq. miles}} \approx 0.438 \text{ million visitors/sq. mile}$$

9. If only $\frac{1}{3}$ of those visitors are in Miami during the winter months, what is the population density of visitors?

$$\frac{8.067 \text{ million}}{55.25 \text{ sq. miles}} = 0.146 \text{ million/sq. mile}$$



10. If $\frac{3}{4}$ of the visitors are in Miami during the summer months, what is the population density of visitors?

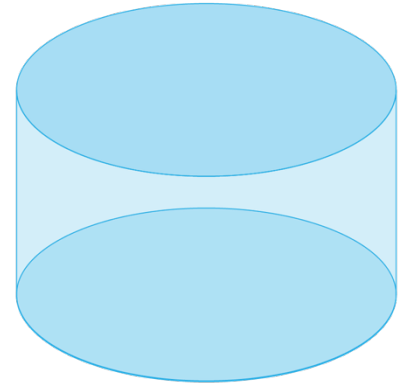
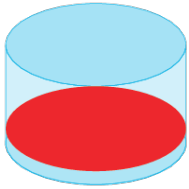
$$\frac{18.15 \text{ million}}{55.25 \text{ sq. miles}} \approx 0.329 \text{ million/sq. mile}$$

Geometry
Unit 9
Lesson 3 Homework

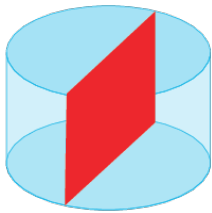
Name Answer Key
Date _____
Period _____

A cylinder is shown.

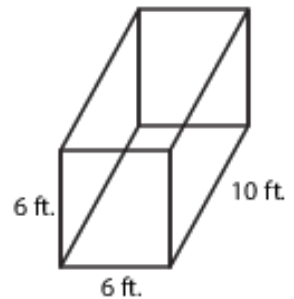
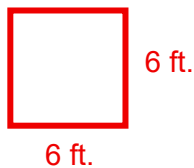
1. Draw a cross section parallel to the base.



2. Draw a cross section perpendicular to the base.

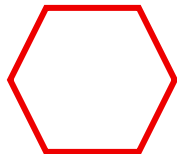


3. A square prism is shown with base edge length of 6 feet (ft.) and a height of 10 ft. Draw the cross section formed by the intersection of the prism and a plane parallel to the base. Label the units.

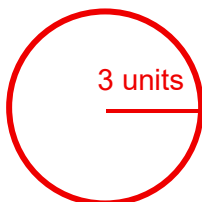


4. Draw a cross section formed by the intersection of a hexagonal prism and a plane that is parallel to the base.

Answers will vary.



5. Draw a cross section formed by the intersection of a sphere with a radius of 3 units and a plane passing through the center of the sphere. Label the units.



6. Draw a cross section formed by the intersection of a triangular prism and a plane perpendicular to its the base.

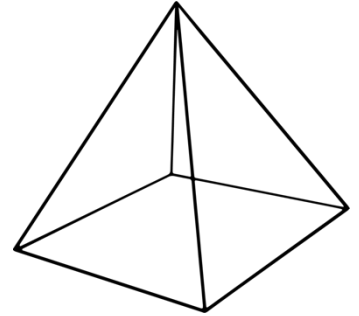
Drawings will vary.



A square pyramid is shown.

7. Describe the shape of a cross section parallel to the base.

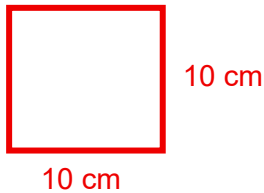
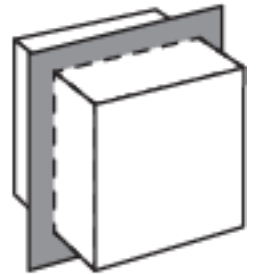
The cross section will be a square.



8. Describe the shape of a cross section perpendicular to the base.

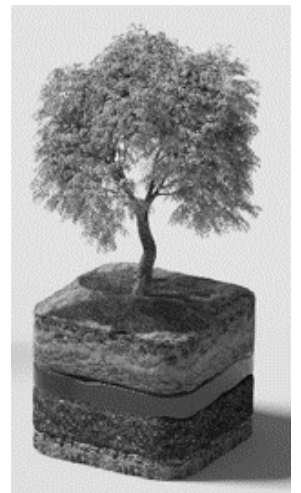
The cross section will be a triangle.

9. A cube is shown, with an edge length of 10 centimeters (cm). Draw the cross section formed by the intersection of the cube and a vertical plane passing through the center of the cube.



10. What two-dimensional shape is created by a horizontal cross section going through the second layer of earth in the picture?

Answers will vary. A square



Geometry
Unit 9
Lesson 4 Homework

Name _____ **Answer Key** _____
Date _____
Period _____

Given the graph.

1. Draw the transformation of the triangle reflected across the y -axis.

See the solid red lines on the graph

2. What three-dimensional shape would be created by rotating the triangle around the y -axis?

Cone

3. What are the base and height measurements of the three-dimensional shape, in units?

Base radius = 4 units

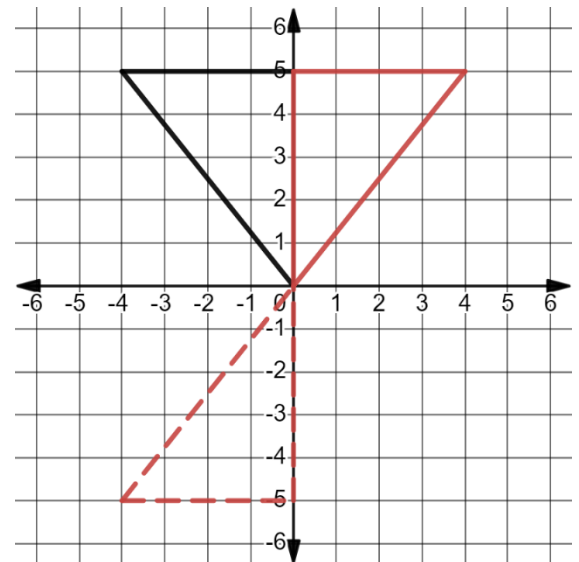
Height = 5 units

4. Draw the transformation of the triangle reflected across the x -axis in a different color.

See the dashed lines on the graph

5. What three-dimensional shape would be created by rotating the triangle around the x -axis?

Cylinder with a cone cutout



Given the circle.

6. Draw the transformation of the circle reflected across the y -axis.

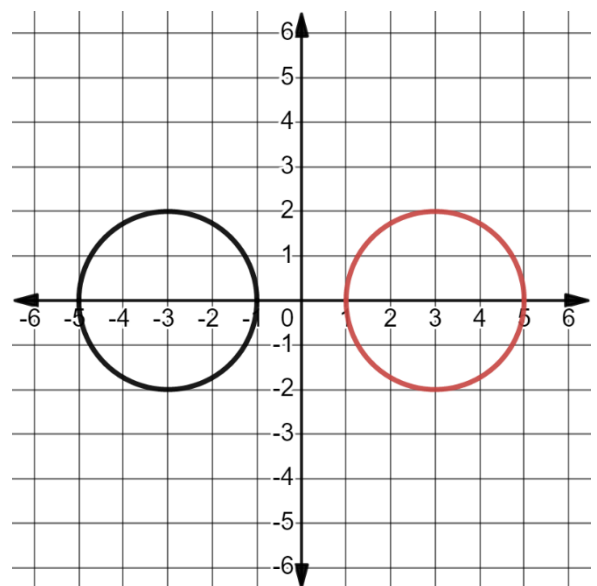
See red circle on graph

7. What three-dimensional shape would be created by rotating the circle around the y -axis?

Torus or donut

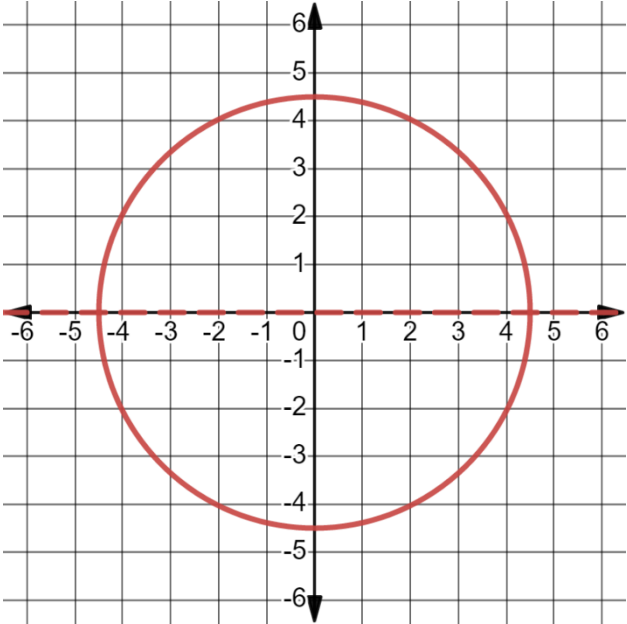
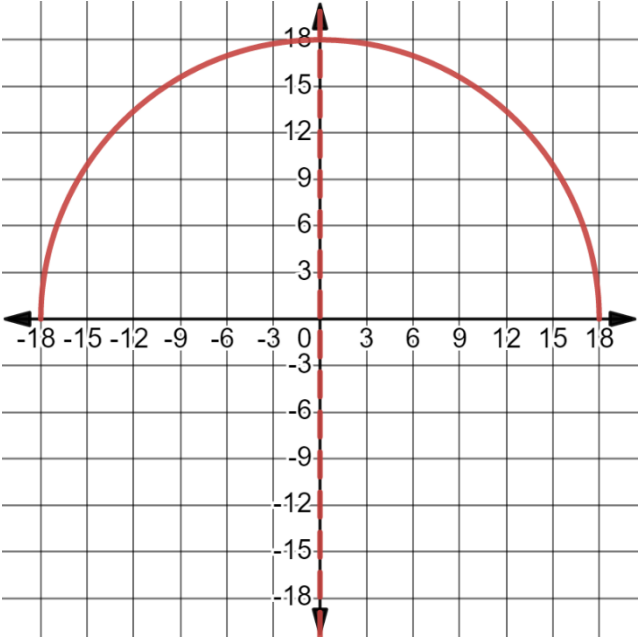
8. Name one real world object that exhibits the three-dimensional shape created.

Answers will vary. Examples: donut, washer

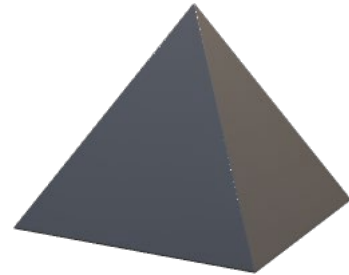


For questions 9-10, draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

Answers will vary.

Image	Two-Dimensional Figure
9. A soccer ball with a radius of 4.5 inches 	 Answers will vary. Rotated around the x-axis
10. An umbrella with a radius of 18 inches 	 Answers will vary. Rotated around the y-axis

Using the given three figures, answer questions 1-3.



1. Which two figures can have the same volume?

The tube and the cylinder

2. Explain how the figures listed in Question 1 can have the same volume.

Explanations may vary, but one explanation is that the cross-sections of the pyramid can't be equal in area to the cross-sections of the other two.

3. What information is required for each figure to determine if the volumes are equal?

Radius and height of each

4. Isaac and Dallas went to an amusement park. They each bought a cheeseburger. Isaac's meat patty came with a square base. Dallas's meat patty had a circular base. Is it possible that they both purchased the same size of cheeseburger?

Yes, if the cross-sections have the same area.



5. Dallas's meat patty was 0.75 inches high with a radius of 4.5 inches. What is the volume?

$\pi(4.5)^2(0.75) \approx 47.69 \text{ in}^3$

6. Isaac's hamburger patty was also 0.75 inches high but with a base whose side length was 4.5 inches. Determine if the two containers have the same volume.

$$0.75(4.5)^2 \approx 15.19 \text{ in}^3$$

Not the same volume

7. Brand A nail polish is sold in rectangular prism jars while Brand B nail polish is sold in cylinder jars. Is it possible for a given quantity of Brand A nail polish and Brand B nail polish to have the same volume?

Yes, if the cross-sections have the same area.



8. A jar of Brand B nail polish is 3 inches tall, with a diameter of 1.25 inches. In a store display, there are a total of 18 jars. What is the volume of nail polish in the display?

$$18 \cdot 3(0.625)^2\pi \approx 66.23 \text{ in}^3$$

9. One jar of Brand A nail polish is 3 inches tall, with a width of 1 inch and a length of 1.25 inches. There are 18 jars in the display. What is the total volume of nail polish in the display?

$$18(3)(1)(1.25) = 67.5 \text{ in}^3$$

10. Compare the answers from Questions 8 and 9.

Brand A has the larger volume.

Geometry

Unit 9

Lesson 6 Homework

Name_____ **Answer Key** _____

Date_____

Period_____

1. Find the volume of a rectangular prism with a width of 7.2 feet, a length of 6.9 feet, and a height of 10.5 feet.

$$V = (7.2)(6.9)(10.5) = 521.64 \text{ ft}^3$$



2. Find the volume of a pyramid whose height is 10.5 feet and whose base is a rectangle with dimensions of 7.2 feet by 6.9 feet.

$$V = \frac{1}{3}(7.2)(6.9)(10.5) = 173.88 \text{ ft}^3$$



3. What is the ratio of the answer from Question 1 to the answer from Question 2?

$$\frac{521.64}{173.88} = 3$$

Ratio is 3:1.

4. Find the volume of a sphere with a diameter of 1.74 meters. Use 3.14 for π . Round to the nearest hundredth.

$$V = \frac{4}{3}\pi(0.87)^3 \approx 2.76 \text{ m}^3$$



5. Find the volume of a hemisphere with a diameter of 1.74 meters. Use 3.14 for π . Round to the nearest hundredth.

$$V = \frac{2}{3}\pi(0.87)^3 \approx 1.38 \text{ m}^3$$



6. What is the relationship between the answers from Questions 4 and 5?

The hemisphere has half the volume of the sphere.

7. Determine the volume of a cone with a radius of 16.2 inches, a height of 5.75 inches, and a slant height of 9.9 inches in terms of π .

$$V = \frac{1}{3}\pi(16.2)^2(5.75)$$

$$V = 503.01\pi \text{ in}^3$$



8. Find the volume of a cylinder with a radius of 16.2 inches and a height of 5.75 inches. Round your answer to the nearest thousandth.

$$V = \pi(16.2)^2(5.75) \approx 4740.758 \text{ in}^3 \text{ (when using the pi button)}$$

9. A composite figure consists of a cylinder and a square pyramid. The height of the cylinder is 58.1 centimeters. The height of the pyramid is 23 centimeters. The base of the square pyramid has a width of 19 centimeters. The base of the pyramid sits on one base of the cylinder with the width of the square base equal to the diameter of the cylinder. Find the volume of the composite figure. Round to the nearest hundredth.

$$\text{Cylinder: } \pi(9.5)^2(58.1) = 16473.02 \text{ (when using the pi button)}$$

$$\text{Pyramid: } \frac{1}{3}(19)(19)(23) = 2767.67$$

$$16473.02 + 2767.67 = 19240.69 \text{ cm}^3$$



10. A composite figure consists of a cylinder and a hemisphere. The height of the cylinder is 32.5 inches. The base of the hemisphere sits on one base of the cylinder, both having a diameter of 14 inches. Find the volume of the composite figure. Use 3.14 for π . Round to the nearest hundredth.

$$\text{Cylinder: } V = \pi(7)^2(32.5) = 5000.45$$

$$\text{Hemisphere: } V = \frac{2}{3}\pi(7)^3 = 718.01$$

$$5000.45 + 718.01 = 5718.46 \text{ in}^3$$



1. Oliver lives by the highway. He noticed several tractor trailers drive by every afternoon. Many of the tractor trailers are tankers. The cylinder on a tanker can be 42.6 feet long with a radius of 8.3 feet. What is the maximum volume of the cylinder on a tanker?

$$V = \pi(8.3)^2(42.6) = 2934.714\pi \approx 9219.68 \text{ ft}^3 \text{ (when using the pi button)}$$



2. Oliver counted 16 tanker trucks one Saturday. If every tanker truck was full, what was the total volume of cargo that passed by that day?

$$9219.68(16) = 147,514.88 \text{ ft}^3$$

3. A baseball has a diameter of 7.5 centimeters. What is the volume of a baseball? Round to the nearest hundredth.

$$V = \frac{4}{3}\pi(3.75)^3$$

$$V \approx 220.89 \text{ cm}^3 \text{ (when using the pi button)}$$



4. The diameter of a softball is 8.89 centimeters. What is the volume of a softball? Round to the nearest hundredth.

$$V = \frac{4}{3}\pi(4.445)^3 \approx 367.88 \text{ cm}^3 \text{ (when using the pi button)}$$

5. What is the difference in volume between a baseball and a softball?
 $\approx 146.99 \text{ cm}^3$

6. Penelope went on a road trip. During the drive, she noticed that some hay bales were rectangular prisms and some were cylindrical. She learned that the first hay bale she saw that was a rectangular prism was 16 inches high by 22 inches wide by 44 inches long. What was the volume of the first hay bale? Round to the nearest hundredth.

$$V = 16(22)(44) = 15,488 \text{ in}^3$$

7. The second hay bale was cylindrical with a 5 foot diameter and a 4 foot width. What was the volume of the second hay bale, in cubic inches? Round to the nearest hundredth.

$$V = \pi(2.5)^2(4) \text{ (feet)}$$

$$V = \pi(30)^2(48) \text{ (inches)}$$

$$\approx 135,716.80 \text{ in}^3 \text{ (when using the pi button)}$$

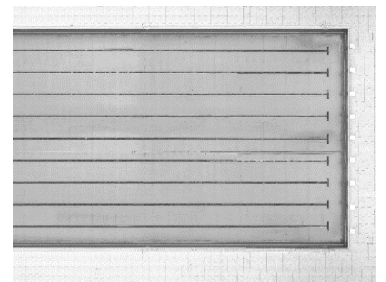


8. Which hay bale is larger?

The cylindrical bale of hay is larger.

9. The high school swimming pool is 25 meters long, 15 meters wide, and 3 meters deep. What is the volume of water in a full pool?

$$V = 25(15)(3) = 1125 \text{ m}^3$$



10. If the pool loses $\frac{1}{10}$ of its water depth during a butterfly swim event, what is the new volume of water?

$$\frac{9}{10}(1125) = 1012.5 \text{ m}^3$$

For questions 1-10, use $\pi = 3.14$ and round all answers to the nearest thousandth if necessary.

1. Find the surface area of a pyramid with a height of 9.8 meters, slant height of 10.34 meters, and a square base with a width of 6.7 meters.

$$(6.7)^2 + \frac{1}{2}(10.34)(26.8) = 44.89 + 138.556 = 183.446 \text{ m}^2$$



2. Find the lateral area of a pyramid whose height is 9.8 meters, slant height is 10.34 meters, and whose base is a square with a width of 6.7 meters.

$$LA = \frac{1}{2}(10.34)(26.8) = 138.556 \text{ m}^2$$

3. Find the lateral area of a rectangular prism with a width of 31.8 feet, a length of 57.1 feet, and a height of 24.1 feet.

$$LA = (177.8)(24.1) = 4284.98 \text{ ft}^2$$

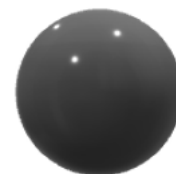


4. Find the surface area of a rectangular prism with a width of 31.8 feet, a length of 57.1 feet, and a height of 24.1 feet.

$$2(31.8)(57.1) + (177.8)(24.1) = 7916.54 \text{ ft}^2$$

5. Find the surface area of a sphere with a diameter of 26.8 centimeters.

$$SA = 4\pi(13.4)^2 = 718.24\pi \approx 2256.418 \text{ cm}^2 \text{ (when using the pi button)}$$



6. Find the surface area of a hemisphere with a diameter of 102.9 millimeters.

$$SA = 3\pi(51.45)^2 = 7941.3075\pi \approx 24,948.353 \text{ mm}^2 \text{ (when using the pi button)}$$

7. The surface area of a sphere is 484π inches. What is the diameter of the sphere?

$$484\pi = 4\pi r^2$$

$$121 = r^2$$

$$11 = r$$

$$d = 22 \text{ in}$$

8. Find the surface area of a cone with a radius of 16.7 yards, a height of 9.14 yards, and a slant height of 19.04 yards.

$$SA = \pi(16.7)^2 + \pi(16.7)(19.04) = 278.89\pi + 317.968\pi = 596.858\pi$$

$$\approx 1875.084 \text{ yd}^2 \text{ (when using the pi button)}$$



9. Find the lateral area of a cylinder with a radius of 6.7 inches and a height of 9.4 inches.

$$LA = 2\pi(6.7)(9.4) = 125.96\pi \approx 395.715 \text{ in}^2 \text{ (when using the pi button)}$$

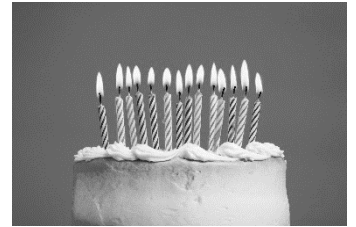


10. Find the surface area of a cylinder with a radius of 6.7 inches and a height of 9.4 inches.

$$SA = 2\pi(6.7)(9.4) + 2\pi(6.7)^2 = 125.96\pi + 89.78\pi = 215.74\pi \approx 677.767 \text{ in}^2 \text{ (when using the pi button)}$$

1. Brea enjoys baking and decorating cakes. The easiest cake pan creates a cake that is 9 inches in diameter and $4\frac{1}{4}$ inches tall. What is the surface area of the cake? Round to the nearest hundredth.

$$SA = 2\pi\left(4\frac{1}{2}\right)\left(4\frac{1}{4}\right) + 2\pi\left(4\frac{1}{2}\right)^2 = 38.25\pi + 40.5\pi \\ = 78.75\pi \approx 247.40 \text{ in}^2 \text{ (when using the pi button)}$$



2. Brea ices every possible space of the cake. She only adds her creative designs to the top of the cake. What is the area she has for her creative designs? Round to the nearest hundredth.

$$A = \pi\left(4\frac{1}{2}\right)^2 = 20.25\pi \approx 63.62 \text{ in}^2 \text{ (when using the pi button)}$$

3. A baseball has a diameter of 7.5 centimeters. What is the surface area of a baseball? Round to the nearest hundredth.

$$SA = 4\pi(3.75)^2 = 56.25\pi \approx 176.71 \text{ cm}^2 \text{ (when using the pi button)}$$



4. The diameter of a softball is 8.89 centimeters. What is the surface area of a softball? Round to the nearest hundredth.

$$SA = 4\pi(4.445)^2 = 79.032\pi \approx 248.29 \text{ cm}^2 \text{ (when using the pi button)}$$

5. What is the difference in surface area between a baseball and a softball?
 $248.29 - 176.71 = 71.58 \text{ cm}^2$

6. Newman's fence needs a new coat of paint. The fence runs along the east side of the property line which is 120 feet long. The height of the fence is 15 feet with a width of 0.5 feet. Both sides of the fence and the top need to be painted. Find the area of the fence that needs to be painted.

$$A = 2(120)(15) + (120)(0.5) = 3600 + 60 = 3660 \text{ ft}^2$$



7. If 1 gallon of paint covers 410 square feet, how much paint does Newman need to buy?

$$3660 \div 410 \approx 8.93$$

Newman needs to buy 9 gallons.

8. If 1 gallon of paint costs \$23.20, how much will the paint cost?

$$9 \times 23.20 = \$208.80$$

9. Gertrude wants to build a new pen for her cow. She has one area of her property available for a rectangular pen that would measure 18 feet by 36 feet. She has a second area of her property available for a square pen that would measure 25 feet wide. Which pen area would require the least amount of fence?

$$\text{Pen 1} = (18 + 36)(2) = 108 \text{ ft}$$

$$\text{Pen 2} = 25 \times 4 = 100 \text{ ft}$$

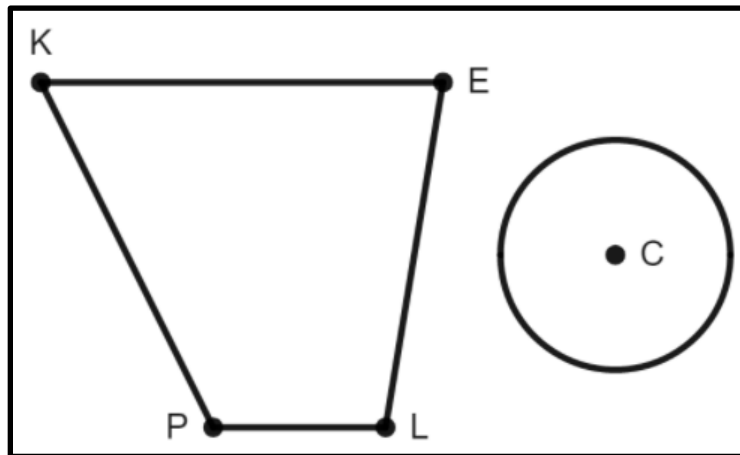
Pen 2 would require the least amount of fence.



10. Gertrude decides to build the pen area with the least amount of fence. If her fence is 7.5 feet tall, what is the surface area of her fence?

$$(7.5)(100) = 750 \text{ ft}^2$$

A diagram with Trapezoid $KELP$ and Circle C is shown. Given the dimensions, find the area of each figure and complete the table. Round to the nearest hundredth if necessary.



Shape	Dimensions	Area
Trapezoid $KELP$	$h = 52.6$, $b_1 = 47.8$, $b_2 = 39.1$	1. $A = \frac{1}{2}(52.6)(47.8 + 39.1)$ $A = 2285.47 \text{ units}^2$
Circle C	$r = 26.7$	2. $A = \pi(26.7)^2$ $A = 2239.61 \text{ units}^2$ (when using the pi button)

Determine the area of the dilated figures and complete the table.

Shape	Area with Scale Factor of $\frac{7}{3}$	Area with Scale Factor of $\frac{3}{7}$
Trapezoid $K'E'L'P'$	3. $2285.47 \left(\frac{7}{3}\right)^2 \approx 12,443.11 \text{ units}^2$	4. $2285.47 \left(\frac{3}{7}\right)^2 \approx 419.78 \text{ units}^2$
Circle C'	5. $2239.61 \left(\frac{7}{3}\right)^2 \approx 12,193.43 \text{ units}^2$	6. $2239.61 \left(\frac{3}{7}\right)^2 \approx 411.36 \text{ units}^2$

7. Parallelogram $CALM$ has a base of 7.6 feet and a height of 21.3 feet. What is the area of Parallelogram $CALM$?
 $(7.6)(21.3) = 161.88 \text{ ft}^2$
8. Parallelogram $CALM$ is multiplied by a scale factor to create Parallelogram $C'A'L'M'$. Parallelogram $C'A'L'M'$ has an area of 485.64 square feet. What is the scale factor for the area of the pre-image to get the image?
 $\frac{485.64}{161.88} = 3$
 $\sqrt{3}$ scale factor of the image
9. Using the scale factor, what is the base of the image?
 $7.6\sqrt{3} \approx 13.16 \text{ ft}$
10. Using the scale factor, what is the height of the image?
 $21.3\sqrt{3} \approx 36.89 \text{ ft}$

For questions 1-10, use $\pi = 3.14$ and round all answers to the nearest hundredth if necessary.

1. When Felix was young, he had a mini trampoline. It was 36 inches in diameter. What was the base (jumping) area of the mini trampoline?
 $r = 18$
 $A = \pi(18)^2 = 324\pi \approx 1017.88 \text{ in}^2$ (when using the pi button)
2. As Felix grew bigger, he needed a bigger trampoline. Felix got a new trampoline for his birthday. It was 128 inches in diameter. What was the base (jumping) area of the new trampoline?
 $r = 64$
 $A = \pi(64)^2 = 4096\pi \approx 12,867.96 \text{ in}^2$ (when using the pi button)
3. What is the difference in areas of the two trampolines?
 $12867.96 - 1017.88 = 11,850.08 \text{ in}^2$
4. With the frame, Felix's new trampoline stood 35 inches high. Felix wants to put a skirt around the trampoline to prevent his pets from going underneath it while he is jumping. What is the lateral surface area of the new trampoline that is to be covered by the skirt?
 $2\pi(64)(35) \approx 14,074.34 \text{ in}^2$ (when using the pi button)
5. Although his pets could not fit under the mini trampoline, what was the lateral surface area of the mini trampoline if it stood 13 inches high?
 $2\pi(18)(13) \approx 1470.27 \text{ in}^2$ (when using the pi button)

6. Hermione is decorating for a birthday party. She has four sizes of gift boxes that are to be placed in one corner, simply for decoration. The base of one is 8 inches by 12 inches, the second is 12 inches by 16 inches, the third is 18 inches by 24 inches, and the fourth is 20 inches by 20 inches. What is the base area of the smallest box?

$$8 \times 12 = 96 \text{ in}^2$$



7. What is the base area of the second box?

$$12 \times 16 = 192 \text{ in}^2$$

8. What is the base area of the third box?

$$18 \times 24 = 432 \text{ in}^2$$

9. What is the base area of the largest box?

$$20 \times 20 = 400 \text{ in}^2$$

10. Hermione designated space in the corner for the gift boxes that is 50 inches by 60 inches. Can she fit all 4 sizes of gift boxes in the designated space?

$$96 + 192 + 432 + 400 = 1120$$

$$1120 < 3000$$

Yes, the total area available is 3000 in^2 and the boxes only take up a total of 1120 in^2 .